

# Divya Swaroopini Shridhar

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## PROFESSIONAL SUMMARY

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Computational Biologist and AI Engineer working at the intersection of bioinformatics, artificial intelligence and biological data science, with experience spanning clinical genomics, infectious disease, transcriptomics and medical AI. Skilled in taking biological problems from raw sequencing and experimental data through computational analysis, workflow automation and biological interpretation. Experienced in building reproducible NGS and metagenomic pipelines, analytical software and AI-driven approaches across Illumina and Nanopore sequencing, clinical variant interpretation and machine learning. Currently extending this work into systems biology, cheminformatics and explainable AI, with a focus on linking computational predictions to underlying biological mechanisms.

## CONFERENCE PRESENTATIONS

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### FUVIBACT: A Novel, Rapid, Sensitive, and Accurate Tool for Infectious Disease Diagnostics

*London Calling 2025 (Oxford Nanopore Technologies)*

*Co-author*

## EXPERIENCE

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### Bioinformatician and AI Consultant

*Dhiti Omics Technologies Private Ltd*

Jan 2025 - Dec 2025

*Bangalore, Karnataka*

- Re-engineered a manual infectious-pathogen analysis workflow that could take 12 hours per case into an automated Nextflow/shell-based pipeline completing analysis in 2 hours, progressively eliminating hours of manual processing in the initial automation stage and ultimately saving 7–8 hours in critical/ICU analyses through automated QC, classification, database retrieval and BLAST-based verification. Integrated Kraken2, Porechop, BLAST and AI-assisted pathogen ranking, with case-specific modifications and curated biological databases, creating a clinically usable workflow accessible to non-programmers.
- Served as the primary bioinformatics analyst for infectious-disease sequencing cases, independently analysing Nanopore and Illumina NGS data across metagenomic, amplicon and trio workflows, while performing QC, alignment, variant calling, structural variant and CNV analysis, annotation and clinical interpretation. Used real clinical cases to identify workflow bottlenecks and iteratively validate each automation stage before incorporating it into the production workflow.
- Developed AI-assisted clinical variant interpretation capabilities, combining automated ACMG-based pathogenicity classification with LLM-driven phenotype-to-gene prioritisation to help reduce large variant sets to clinically relevant candidates. Built and maintained 7+ curated, target-specific biological databases supporting genomic analysis and predictive-health workflows, while contributing to the development of FUVIBACT, subsequently presented at London Calling 2025.

### Developer and Data Analyst Intern

*Centre for Cellular and Molecular Platforms (BLiSC)*

May 2022 - Nov 2022

*Bangalore, Karnataka*

- Worked at the intersection of microfluidics, cell sorting and data analytics, learning the experimental workflow behind a microfluidic cell-sorting device and translating raw experimental measurements into interpretable analytical outputs for device development.
- Developed a web-based visualisation and analysis platform for fluorescence time-series data, replacing repetitive manual analysis with an interactive workflow that enabled faster comparison of experimental runs and supported interpretation of multiple device observations.
- Bridged experimental biology and computational analysis, moving from understanding microfluidic cell-sorting and fluorescence measurement principles to applying statistical signal processing and software-based visualisation to identify performance patterns, anomalies and optimisation opportunities.

## EDUCATION

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### Queen Mary University of London

*M.S. in AI in the Biosciences*

London, UK

*Sep 2023 – Sep 2024*

### Atria Institute of Technology

*B.E. in Information Science and Technology*

Bangalore, Karnataka

*Aug 2018 – Aug 2022*

## PROJECTS

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### Prediction of Oxidative Stress–Mediated Toxicity with Pathway-Level Explainability

2026

*Python · RDKit · Cheminformatics · Transcriptomics · Explainable AI · SHAP · Pathway/Gene-Set Analysis · Mechanism-Aware ML · Agentic AI · Computational Toxicology*

- Developing a mechanism-aware ML framework integrating chemical structure and transcriptomic features to predict compound-induced oxidative stress and potential oxidative burden.
- Applied RDKit cheminformatics for molecular standardisation, physicochemical descriptors, molecular fingerprints and identification of redox-reactive structural motifs.
- Integrated transcriptomic signatures with ROS/NRF2 pathways and gene modules, using pathway enrichment and SHAP explainability to connect predictions to biological mechanisms; developing an Agentic AI evidence-integration framework.

### Intelligent Variant Interpretation Pipeline

2025

*Illumina NGS · ACMG Guidelines · Variant Annotation · Variant Prioritisation · Phenotype-to-Gene Mapping · LLMs · Clinical Data Mining · Precision Medicine · Pipeline Development*

- Built and benchmark-tested an end-to-end Illumina variant interpretation pipeline, automating variant processing and ACMG-based pathogenicity classification for clinical variant triage.
- Explored LLM-driven phenotype-to-gene prioritisation, using patient clinical histories to identify the relevant genes and add clinical context beyond conventional variant annotation.
- Integrated genomic evidence with AI-assisted clinical reasoning, developing approaches for variant prioritisation, phenotype-to-gene mapping and literature-based evidence analysis.

### Comparing Speech Recognition Modules for the Identification of Medicine Names

2024

*OpenAI Whisper · Hugging Face · LLaMA-3 70B · QLoRA · Tortoise TTS · Flask · NLP · ASR · Model Fine-Tuning · Synthetic Data · WER/CER/PER*

- Developed a Whisper-based ASR system for medication-name recognition across diverse UK accents, combining public dialect datasets with 100 AI-generated medicine-name audio samples.
- Compared Whisper Medium, Whisper + LLaMA-3 70B and QLoRA fine-tuned Whisper using WER, CER and PER, with fine-tuning performed over 4,000 training steps.
- Reduced noisy-condition WER from 1.5 to 1.3 and supported deployment through a Flask-based real-time transcription application.

### EndoNet: Computationally Efficient AI for Capsule Endoscopy

2024

*PyTorch · CNNs · NVIDIA CUDA/GPU · Augmentor · Deep Learning · Medical Computer Vision · Image Classification · Imbalanced Data · Data Augmentation · Model Evaluation*

- Designed and trained a lightweight 3-layer CNN for 14-class gastrointestinal image classification, using LeakyReLU, dropout and combined max/average pooling for computational efficiency.
- Addressed severe class imbalance across 47K+ capsule endoscopy images using targeted down-sampling, class-specific augmentation and weighted sampling.
- Achieved 90.65% test accuracy and 0.9205 F1-score, with training at 2.5 min/epoch on an NVIDIA RTX 4050, demonstrating an efficient alternative to heavier architectures.

## TECHNICAL SKILLS

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**Bioinformatics, Genomics & Data Science:** Nextflow, GATK, BWA, Bowtie2, SAMtools, FastQC, MultiQC, Kraken2, BLAST, Porechop, EPI2ME, DRAGEN, VariantStudio, Illumina, Nanopore, NGS, Variant Calling, Structural Variant Detection, CNV Analysis, Variant Annotation, ACMG Interpretation, Metagenomics, Amplicon Sequencing, Transcriptomics, RNA-seq, Differential Expression, Gene-Set & Pathway Analysis, Cheminformatics, RDKit, GC-MS Metabolomics, Statistical Analysis, Biomarker Discovery, Machine Learning, Deep Learning, Explainable AI, SHAP, NLP, Computer Vision, CNNs, PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, LLMs, Agentic AI

**Programming, Software & Cloud:** Python, R, SQL, Java, JavaScript, Dart, Julia, Groovy, Flask, REST APIs, Flutter, PyQt, AWS, Linux, Git, Docker, Workflow Automation, Data Processing, Model Deployment